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A Review of Rate Coefficients in the H_2-F_2 Chemical Laser System

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measured k_{1a} and k_{1b} in a flow system using an isothermal catalytic probe over the temperature range of 77 to 300 K and found temperature dependencies of $T^{-0.81}$ and approximately $T^{-0.61}$, respectively, with values at 300 K of 3.3 and 2.9×10^{15} cc^2/mol^2-sec. Their results, extrapolated to the temperature range of the experiments of Jacobs, Giedt, and Cohen (1967), agree with the results of the latter workers within 35%. Because the lower temperatures are of more interest in laser chemistry, the results of Ham, Trainor, and Kaufman (1970) are weighted heavily. The expressions $k_{1a} = 6.2 \times 10^{17} T^{-0.95}$ and $k_{1b} = 9.4 \times 10^{16} T^{-0.61}$ fit both high- and low-temperature data within about 10%. The numerous results are not tabulated here; further citations are given by Baulch (1972).

Of the several room-temperature measurements of k_{1c}, the most reliable seems to be that of Bennett and Blackmore (1968). They obtained an upper limit of 2.5×10^{15} cc^2/mol^2-sec in a discharge flow system using esr for atom detection, suggesting that k_{1c} has a maximum value somewhere in the temperature range 1000 to 3000 K and is relatively unimportant at low temperatures. The expression $k_{1c} = 1.2 \times 10^{14} T^{1/2}$ is consistent with the upper limit of Bennett and Blackmore (1968) and agrees with the low-temperature end of the shock-tube experiments of Jacobs, Giedt, and Cohen (1967). Therefore, it is probably a reasonable expression in the temperature range of interest, although it is not useful above 3000 K. This expression is recommended in Appendix I, subject to modification when more experimental data become available.

B. F_2 DISSOCIATION

A thorough critical review of the dissociation-recombination of F_2 and Cl_2 was published by Lloyd (1971); repetition of his work has been kept to a minimum here.

In recent years, several shock-tube determinations have been made of the rate of F_2 dissociation (Table 1).

$$F_2 + M \rightleftharpoons F + F + M \qquad (2)$$

Table 1. Fluorine Dissociation Rate Coefficient

$$F_2 + M \xrightarrow{k_2} 2F + M$$

$$k_2 = A \exp(-E/RT)$$

M	T, K	$P_{HF:M}$ atm-cm	k_2 cc/mole-sec	Reference
HF	140	0.014	2×10^{12}	Azen and Freed (1971)
	1400-4100	$6.1 \times 10^{-4} \exp(4/T^{1/3})$	$6.6 \times 10^{2} T^{2.44}$	Solomon et al. (1971)
	1150-4000	$1.0 \times 10^{-2} \exp(14.4/T^{1/3})$	$1.1 \times 10^{5} T^{2.12}$	Bott and Cohen (1971)
	140	0.025	1.1×10^{12}	Stephens and Cool (1972)
	294	0.015	1.6×10^{12}	Hancock and Green (1972)
	295-1000	$\exp(-41/T^{1/3})$	$(1 \times 10^{14} T^{-1} + 1.7 \times 10^{5} T^{2})$	Bott and Cohen (1972)
	240	0.025	9.5×10^{11}	Osgood, Jayan, and Sanders, (1972, 1974)
	150-750	$1.9 \exp(-29/T^{1/3})$	$(2.6 \times 10^{14} T^{-1} + 7.1 \times 10^{8} T)$	Freed, Wilson, and Taylor (1973)
	295-1000	$8.0 \exp(-38.9/T^{1/3})$	$(1 \times 10^{14} T^{-1} + 1.5 \times 10^{5} T^{2})$	Hinchen (1973a, 1973b)
	297-670	$5.7 \exp(-39/T^{1/3})$	$(4.2 \times 10^{14} T^{-1} + 1.5 \times 10^{5} T^{2})$	Lacks and Child (1974)
	1500-5000	$5.7 \times 10^{-1} \exp(42.0/T^{1/3})$	$2.8 \times 10^{8} T^{2.1}$	Vasil'ev et al. (1973)
	600-2400	$15.5 \times 10^{6} T^{-2} + 4.4 \times 10^{-1} T^{-1}$	$(4.4 \times 10^{14} T^{-1} + 4 \times 10^{5} T^{2})$	Blair, Brashears, and Schott (1973)
	1400-2600	$0.026 \exp(21.4/T^{1/3})$	$1.6 \times 10^{6} T^{1.78}$	Yost and Rimpel (1972)
	100	0.022	1.1×10^{12}	Abt and Cool (1971)
	150	0.027	0.9×10^{12}	Abt and Cool (1971)
Ar	1150-4000	$1.6 \times 10^{-3} \exp(112/T^{1/3})$	$8.5 \times 10^{-4} T^{4.01}$	Bott and Cohen (1971)
	1500-5000	$8.5 \times 10^{-1} \exp(99.47/T^{1/3})$	$0.06 T^{1.41}$	Vasil'ev et al. (1973)
	800-2400	$[90 \exp(-85/T^{1/3}) + 10^{-2}]^{-1}$	$1.7 T^{1.05}$	Blair, Brashears, and Schott (1973)
	150	>4	$< 8 \times 10^{9}$	Azen and Freed (1971)
	295	>22	$< 1.3 \times 10^{9}$	Hinchen (1973a)
	295	>22	1.1×10^{9}	Hancock and Green (1972a)
He	1150-4000	$1.9 \times 10^{-4} \exp(133/T^{1/3})$	$5.5 \times 10^{-3} T^{4.46}$	Bott and Cohen (1971)
	295	>22	$< 1.1 \times 10^{9}$	Hinchen (1973a)
F	1400-4100	$1.4 \times 10^{-5} \exp(4/T^{1/3})$	$1.2 \times 10^{4} T^{2.85}$	Solomon et al. (1971)
	1900-3300	$2.5 \times 10^{-6} T$	1.2×10^{11}	Bott and Cohen (1971b)
	2045	0.026	8×10^{12}	Bauer and Solomon (1971)
	1400-2400	$1.4 \times 10^{-4} \exp(4.4/T^{1/3})$	$6.3 \times 10^{4} T^{2.40}$	Blair, Brashears, and Schott (1973)
	295	0.014	1.7×10^{11}	Quigley and Wulge (1974)
F_2	140	>1	2.5×10^{10}	Azen and Freed (1971)
	150	>11	$< 2 \times 10^{9}$	Freed, Wilson, and Taylor (1973)
H_2	295-810	$5.3 \times 10^{-2} T^{-0.28}$	$1.6 \times 10^{4} T^{2.28}$	Bott (1974)
	1400-4000	$0.26 \exp(4.4/T^{1/3})$	$4.4 \times 10^{4} T^{2.12}$	Bott and Cohen (1972)

Although the numerical values of the rate coefficient have generally been in agreement, within the experimental uncertainties of the various determinations, the temperature dependencies of the rate coefficient have been noticeably inconsistent. The various results are summarized in Table 1. The rate coefficients calculated from shock-tube data are plotted (Fig. 1); the recombination rate coefficients have been calculated and extrapolated to lower temperatures (Fig. 2). The extent to which the F_2 bond energy exceeds the average experimental activation energy is disconcerting. This discrepancy becomes important when calculating the rate coefficient for F + F recombination from $K_{4,-4}$ and k_4 and extrapolating the results to low temperatures. The problem is compounded by the uncertainty in the bond dissociation energy of F_2.

Although a critical review of the bond strength of F_2 is outside the scope of this report, it is evident that the possible uncertainty in such a fundamental datum leads to ambiguities in several thermochemical and kinetic values important to this study. In 1968, a National Bureau of Standards committee recommended 37.76 kcal/mol for $D^\circ_{298}(F_2)$; since 1965, the JANAF committee has recommended 37.72 ± 0.8 kcal/mol. (The dissociation energy at 0 K, D°_0, in either case is 1.0 kcal/mol smaller.) Here, 38 kcal/mol is used; in view of the uncertainties, a third significant figure is unnecessary.

With a value of 38 kcal/mol for $D^\circ_{298}(F_2)$, all of the shock-tube values for the activation energy of F_2 dissociation in an Ar bath lead to a strongly negative temperature dependence for F-atom recombination, which results in unreasonably large values near 300 K. Therefore, for calculations, we prefer the theoretical value calculated by Shui, Appleton, and Keck (1971). Their results agree numerically with the shock-tube data of Johnson and Britton (1964) but lead to an activation energy of 35.1 kcal/mol and a pre-exponential factor of $10^{13.7}$. Until direct measurements near 300 K are available, $k_4^{Ar} = 10^{13.7} \exp(-35,100/RT)$ is recommended, which implies

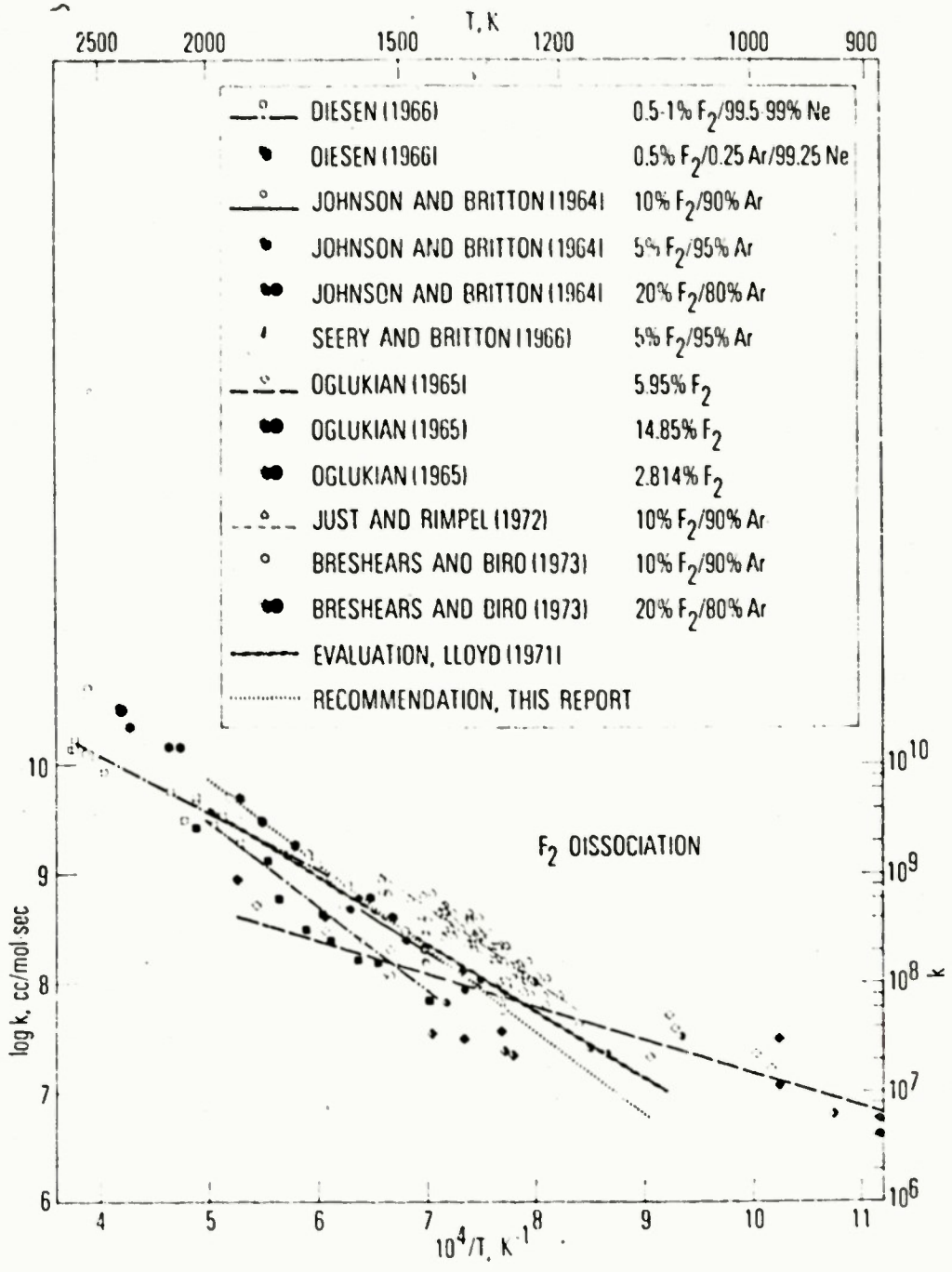


Fig. 1. Experimental Measurements of F_2 Dissociation Rate Coefficient k_1 Versus $10^3/T$

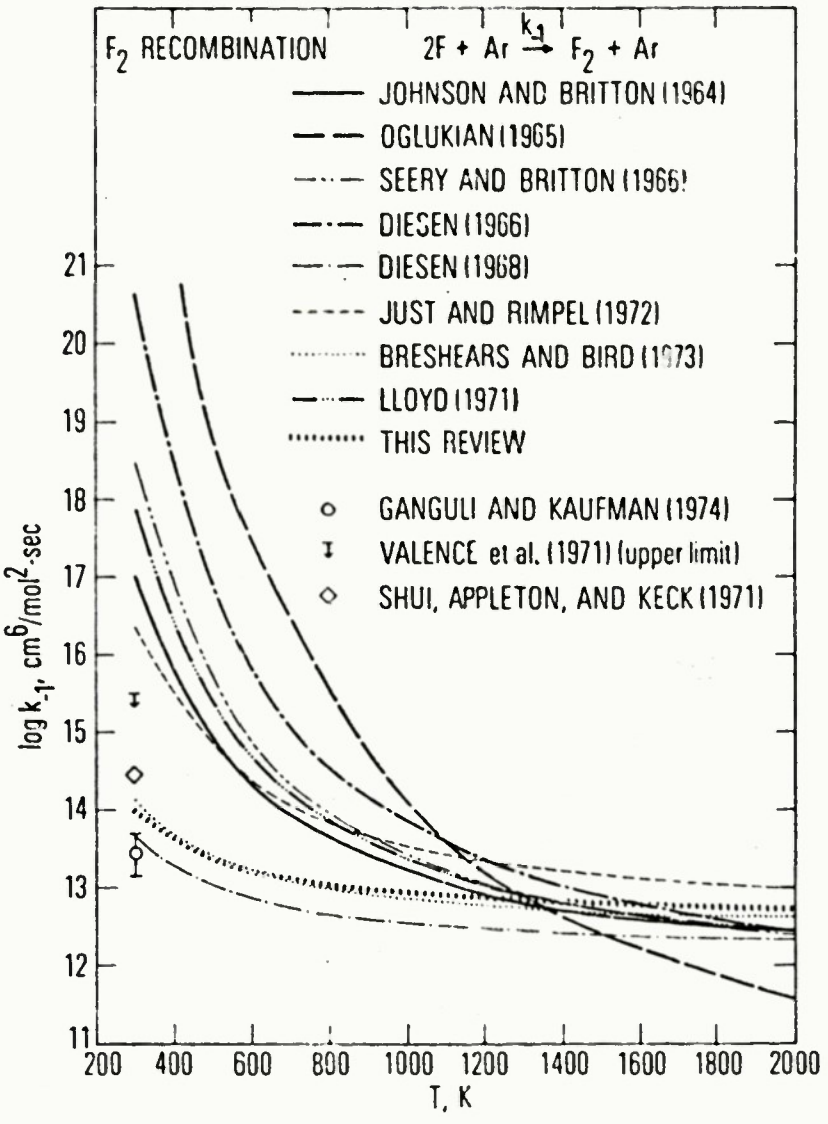


Fig. 2. F + F Recombination Rate Coefficient k_{-1} Computed From k_1 or Measured Directly Versus T

that k_{-4}^{Ar} at 300 K is $10^{14.0}$. A recent report by Ganguli and Kaufman (1974) suggests log $k_{-4}^{Ar} \approx 13.5$ at 300 K, which is smaller than the recommended value for k_{-4} by a factor of ≤3.

As a result of their recent work, Breshears and Bird (1973) have concluded that k_4^{F2}/k_4^{Ar} is approximately 2.7 in the temperature range 1400 to 2600 K. However, an examination of Fig. 1, in which all the shock-tube dissociation data are plotted, shows that their data are not greatly sensitive to this ratio; therefore, considerable uncertainty must be assigned to this value. Data on the rate of F_2 dissociation with M = H, H_2, or F are not available. From an analogy with the findings in cases of other homonuclear diatomics such as Cl_2, Br_2, H_2, O_2, and N_2, it is expected that F atoms will exhibit a large efficiency relative to Ar; $k_4^F = 10\ k_4^{Ar}$ is assumed in this report.

If it is assumed that the primary recombination mechanism for F atoms is, as is believed to be the case for I atoms, the radical-molecule mechanism

$$F + M \rightarrow FM^*$$

$$FM^* + F \rightarrow F_2 + M$$

then some qualitative statements can be made regarding the relative efficiencies of H and HF as chaperones. For example, H atoms would be expected to be somewhat more efficient than Ar because the strong H-F bond would favor formation of FM*, where M = H. The effect on HF chaperone efficiency would be much less pronounced. Therefore, in this report, it is tentatively assumed that H atoms are three times as efficient as Ar and that HF is equal to Ar in efficiency. Other species are assumed to have the same efficiency as Ar.

Figure 2 is significant in demonstrating that a relatively small discrepancy in k at shock-tube temperatures can lead to an unacceptably large